



Project Title	Rising Waters: Machine Learning Based Flood Prediction System		
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14. Functional Features

The feature list below is drawn directly from what the application actually does end-to-end, from the README's stated feature set through to the concrete logic in app.py. Each feature is grouped loosely by whether it lives in the ML layer, the backend, or the frontend, though several — like severity classification — span more than one layer.

Feature	Description
Multi-model comparison	Automatically trains and compares Decision Tree, Random Forest, KNN, and XGBoost, deploying the best performer.
Instant weather analysis	Users receive a flood probability score immediately after submitting input data.
Smart Auto-Fill	A test-data generator instantly populates the form with realistic values to demonstrate the tool without manual entry.
Severity classification	Results are classified into Normal, Elevated Flood Watch, or Severe Flood Warning tiers.
Contextual insights	Dynamic, human-readable explanations (e.g. excessive rainfall, high humidity) accompany the numeric prediction.
Recommended protocols	Actionable safety guidance is generated based on the severity tier.
Responsive glassmorphism UI	A fully responsive, modern interface built with vanilla HTML/CSS/JS.

Of these, Smart Auto-Fill and Contextual Insights are the two features that most directly address the usability gap identified back in the Empathy Map and Problem-Solution Fit sections: Auto-Fill removes the barrier of needing exact figures on hand, while Contextual Insights removes the barrier of needing to interpret a raw probability unaided.

